## (a) Privileged teacher supervision

**Simulation ground truth**

Object pose & velocity

Goal orientation & error

**Proprioception**

Hand state + prev. action

**Teacher network**

Actor MLP

Critic MLP

Teacher action

**Simulator**

Reward / PPO

**Frozen teacher**

Forward

Supervision

## (b) Distill a goal-conditioned student

Language

**Semantic agent**

Public skill selection

SigLIP2 skill bank

**Goal-bank vector**

g

Sampled during training

Goal projection Linear + tanh

z_g

**RGB observation**

Observed RGB (simulation / camera)

Frozen pose CNN

Pose encoder MLP

Training input: noisy sim pose

Hand state

Proprioception + history

**Concat**

**Student policy**

Recurrent encoder

Actor MLP

Action a_S

**Distillation**

Action MSE

State MSE (training)

Training only

State decoder MLP

**Deploy student**

## (c) Closed-loop execution

Live agent

Student policy

Wait / move controllers

Hand actions

**Physical hand**

TODO: physical setup
No hardware result shown

**Target occluded**

TODO: physical inspection frame

**Target presentation**

TODO: physical target frame

**Language instruction**

Present the face marked T

New RGB